



TMV Monitoring Device Hardware & Firmware Technical Specification

Nube iO ACX platform — Galvin CliniMix-TMV (Generation 2)

Device-level reverse specification — Nube iO Hardware & Firmware Team

Document	Nube iO ACX (ACX-001) — CliniMix-TMV Hardware & Firmware Specification
Prepared by	Nube iO — Hardware & Firmware Team
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Responds to	Galvin Engineering — Temperature Monitoring Gen 2 Design Scope, Doc 004.00.00.30 Rev 0 (NPD/PDDR 1442N)
Classification	Commercial in Confidence

1. Scope & Purpose

This document is the **Nube iO Hardware & Firmware Technical Specification** for the TMV Monitoring Device — the Nube iO **ACX** platform (drawing ACX-001) — that Nube iO is developing for Galvin Engineering's **CliniMix-TMV / CliniMix Generation 2** water-safety platform.

It is a *reverse specification*: it restates, from the hardware and firmware team's point of view, only the parts of Galvin's design intent that Nube iO is responsible for building, and expands each one with the engineering detail required to take the product from first concept to a manufacturable design. Galvin's document is a system-level design scope; this document is the device-level response to it.

Primary input. Galvin Engineering, *Temperature Monitoring Gen 2 — Design Scope*, Document No. 004.00.00.30, Revision 0, NPD/PDDR 1442N (hereafter "the Design Scope").

What Nube iO owns (this document): the monitoring transmitter — its electronics, enclosure interfaces, power architecture, on-device measurement and event logic, connectivity, security and OTA, and the local control of the optional solenoid output.

What Nube iO does not own (interface only): the cloud platform, the CliniMix Management Portal, the dashboard, email/alert generation, group/threshold administration and the maintenance-date ledger. Galvin's Design Scope marks these explicitly as **cloud logic**; the device is deliberately constrained to a *telemetry and execution node* (Design Scope §6.1.2). This specification treats every cloud-owned parameter as a value the device stores and applies, not one it decides.

This is Nube iO's **first design pass** against the Gen 2 scope. Where the Design Scope records an item as *TBD*, this document carries it into **§7 Open Items & Engineering Recommendations** with a proposed resolution rather than leaving it open.

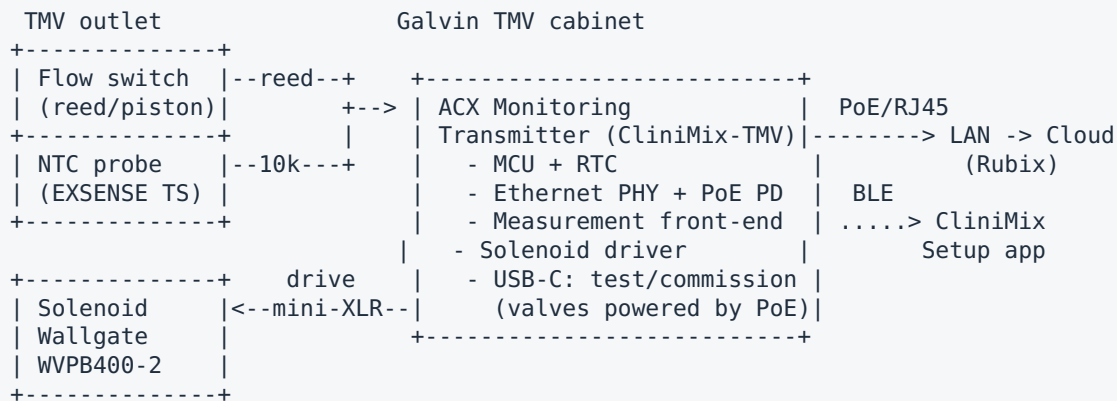
1.1 Reference documents

Tag	Document	Identifier
[DS]	Galvin Engineering — <i>Temperature Monitoring Gen 2 Design Scope</i>	Doc 004.00.00.30 Rev 0 / NPD/PDDR 1442N
[TS]	EXSENSE Sensor Technology — <i>TS Series NTC Temperature Sensor, Specifications Approval Sheet</i> (customer: Galvin Engineering)	P/N TS103F25C3950FA-ML1300A · SAS21022503W · 2021-02-25
[WG]	Wallgate — <i>Installation Drawing for WVPB400-2 (Solenoid)</i>	WVPB400-2
[PO]	Nube iO — <i>CliniMix-TMV Product Scope (Product Overview)</i> , prepared for Galvin Engineering	Nube iO drawing ACX-001 Rev V01
[FL]	Nube iO ↔ Galvin — <i>Hardware Development Feedback Loop (TMV-ACX)</i>	live issues log, 9 items

The Product Overview [PO] supersedes the earlier iO8G Executive Summary as the Nube iO product reference. Cross-references below are written as, e.g., (*DS §4.1*) for the Design Scope, (*TS §3*) for the thermistor datasheet, (*WG*) for the solenoid drawing, (*PO*) for the Product Overview, and (*FL 1.0*) for a Feedback Loop item.

2. System Context

The device is a **direct device-to-cloud** transmitter mounted inside the Galvin TMV cabinet. It takes its sole network path from a single Ethernet/PoE drop to the building network, measures outlet temperature and flow at the valve, computes per-event summaries on-device, and pushes them to the Galvin/Rubix cloud. There are no on-site controllers or gateways in the data path (*DS §3.1*, *§4.7*).



Roles relevant to the device (DS §6.1.4): the plumber commissions it over BLE with the CliniMix Setup app; the facility manager and Galvin manage thresholds and alerts entirely in the cloud. The device exposes no general-user control surface beyond the momentary service button.

3. Hardware Specification

3.1 General electrical & environmental

Parameter	Specification	Source
Operating ambient temperature	0 °C to 65 °C	DS §4.1
Ingress protection	IPx4 required (DS); ACX platform rated IPx5	DS §4.1, §4.3; PO
Primary power	Power over Ethernet (PoE), 100 Mb single-cable; PoE class TBD per device budget — must also power the valve(s) — see §7	DS §4.1; PO; FL 7.0
Secondary power	USB-C — test and commissioning only ; 12 VDC jack is the alternative field supply on the ACX platform	DS §4.1, §5.5.3; PO; FL 8.0
Concurrent-supply protection	Circuit protection must permit USB and PoE to be connected simultaneously without damage	DS §4.1
Real-time clock	On-board RTC, disciplined to cloud time; all timestamps synchronised to cloud	DS §4.1
Time zone	Configurable (cloud-pushed)	DS §4.1

3.2 Core components

Building on the established Nube iO **ACX** platform (PO, drawing ACX-001 Rev V01) — which provides 8 × I/O (2 RO, 2 AO 0–10 VDC, 4 UI), PoE/12 VDC power, an onboard RTC and optional RS-485/LoRa — the transmitter is built around:

- **MCU:** application microcontroller providing **BLE** for the CliniMix Setup app and **optional Wi-Fi** (DS §4.4). Primary connectivity remains wired LAN.
- **Ethernet PHY + PoE PD front-end:** 100 Mb Ethernet (PO) with an IEEE 802.3 PoE powered-device controller. Final PoE class is set by the power budget, which must include the valve load (§7; FL 7.0).
- **Real-time clock** with battery/supercap hold-over for timestamp continuity across power loss (DS §4.1; PO).
- **Measurement front-end** for the NTC probe and reed flow switch (§3.6, §3.7).
- **Solenoid driver** stage, valve powered from PoE with USB-C reserved for test/commissioning (§3.8; FL 7.0/8.0).

3.3 Power architecture

The device supports two independent supply paths with seamless, automatic hand-over (DS §4.1, §5.5.3):

1. **PoE (primary)** — powers the full device, **and the solenoid valve(s)**, over the single 100 Mb RJ45 drop. The feedback loop confirms PoE must supply the valves (FL 7.0); this is the sizing-critical case and refines the Design Scope's original USB-C-fed-solenoid assumption (DS §5.5.2).
2. **USB-C (secondary)** — used for **testing and commissioning only** (e.g. powering the device from a USB power bank before the network drop is live, DS §3.2; FL 8.0). It must still carry enough power to actuate the valve during commissioning checks.

Where USB-C is used it supports **USB Power Delivery (USB-PD)** negotiation to request a source voltage matching the configured solenoid requirement, e.g. 9 V / 12 V or higher (DS §5.5.2). Local energy buffering decouples solenoid in-rush from the supply so activation does not impose abrupt load transients (DS §5.5.2). Power-path management ensures no user intervention is required when switching between PoE and USB-C (DS §5.5.3). Confirming the PoE budget covers device + valve is the highest-priority open item (§7; FL 7.0/8.0).

3.4 Connectors & interfaces

Interface	Connector	Qty	Source
Networking	RJ45 — waterproofing approach under selection (external port, IP67 connector, sealed cable, or internal-connect)	1	DS §4.2; FL 3.0
Power / programming	USB-C	1	DS §4.2
Temperature probe	2-pin IP65 waterproof plug (candidate; pending dev)	1	DS §4.2; FL 2.0
Solenoid output	Male 3-pin mini-XLR , mating the valve's female mini-XLR	1	DS §4.2; FL 1.0
GPIO	TBD	2	DS §4.2
Expansion port	RJ11 — remote I/O cards over BACnet / Modbus	1	DS §4.2, §5.5.4; FL 5.0

3.5 Enclosure, indicators & labelling

Item	Specification	Source
Dimensions (W × L × H)	60 × 120 × 40 mm	DS §4.3
Material	ABS	DS §4.3
Indicators	1 × Power (red), 1 × Status (green) LED	DS §4.3
Button	Momentary status/service button, IPx4	DS §4.3
Colour	Galvin Blue (Pantone PMS-2935), second colour TBD	DS §4.3
Compliance markings	RCM; ABS / e-waste / IP-rating / lithium-battery (if fitted) marks	DS §4.3
Labelling	Galvin Engineering logo, product code/name, I/O designations, QR code	DS §4.3

The QR code carries the immutable manufacturer GUID used to bind the device to the platform at commissioning (DS §4.3, §6.1.3).

The Design Scope enclosure (60 × 120 × 40 mm ABS, IPx4) and the ACX platform reference enclosure (max 150 × 40 × 35 mm, IPx5, per drawing ACX-001 Rev V01, PO) differ; the feedback loop flags Galvin's **detailed box dimensions and internal component layout** as a High-priority input still needed to finalise the casing and connector placement (FL 4.0; §7).

3.6 Temperature-measurement sub-system

Outlet temperature is measured with the **EXSENSE TS-series NTC probe already specified to Galvin (TS)**, read only while the flow switch indicates flow (DS §5.5.5). The measurement front-end and the probe together set the accuracy budget.

Probe — EXSENSE P/N TS103F25C3950FA-ML1300A (TS §1–§3):

Characteristic	Value	Source
Element	10 kΩ NTC thermistor	TS §3-1
Resistance @ 25 °C (R25)	10 kΩ ± 1 %	TS §3-1
B-value B(25/50)	3950 K ± 1 %	TS §3-2
Thermal time constant τ	≤ 12 s	TS §3-3
Dissipation factor δ	≥ 0.8 mW/°C	TS §3-4
Rated power	5 mW max	TS §3-7
Housing / thread	Stainless steel 304, NPT 1/8	TS §1
Seal	Silicone O-ring	TS §1

Characteristic	Value	Source
Lead	UL4478, #24 AWG, 1300 mm	TS §1
Insulation / withstand	$\geq 100 \text{ M}\Omega$ @ DC 500 V; AC 1200 V, 2 mA, 1 s	TS §3-5/3-6

Design implications for the Nube iO front-end:

- The Design Scope mandates **internal measurement-circuitry tolerance $< 1 \%$** (DS §5.5.5). Combined with the probe's $\pm 1 \%$ R25 / $\pm 1 \%$ B-value (TS §3-1/3-2), the bias network and ADC reference must be selected so the *electronics* contribution stays inside that 1 % budget.
- Over the clinically relevant 40–60 °C mixed-water band the probe presents roughly **5.34 k Ω $\rightarrow$ 2.48 k Ω** (TS R-T table, §10 below); the bias divider and ADC range should be centred on this band for best resolution while still resolving the full 0–65 °C ambit.
- Self-heating must be limited: at 5 mW rating and $\delta \geq 0.8 \text{ mW}/^\circ\text{C}$ (TS §3-4/3-7), sense current is chosen to keep self-heating error well under the 1 % circuitry budget.
- $\tau \leq 12 \text{ s}$ (TS §3-3) bounds how quickly the probe tracks a step change; this informs the on-device sampling/averaging window during a flow event (§4.2) and supports the Design Scope's 5 s post-stabilisation compliance reading (DS §4.6).

3.7 Flow-switch input

Flow presence is sensed by the TMV outlet's mechanical **reed flow switch** (DS §5.5.5). With no flow the reed contact is open; flowing water drives the magnetic piston, compresses the spring, closes the reed contact and so closes the thermistor measurement circuit (DS §5.5.5). The input must:

- Detect "Flow" when the switch is active for $\geq 1 \text{ s}$ (DS §5.1.3).
- Apply **debounce in hardware or firmware** to reject reed contact bounce (DS §5.5.5, §3.7).
- Gate thermistor sampling on flow state so temperature is only logged during genuine flow (DS §5.5.5).

3.8 Solenoid driver output

The device provides a dedicated output to drive an **external DC solenoid valve** — the reference valve being the **Wallgate WVPB400-2 (WG)**, an inline brass compression-fitting solenoid supplied with a flying lead. The valve terminates in a **female mini-XLR**, so the device side uses a mating **male 3-pin mini-XLR** connector (FL 1.0; cf. Wallgate WDC-100-NX cable). Driver requirements (DS §5.5.2):

Requirement	Specification	Source
Supported solenoid supply	6 V to 24 V DC	DS §5.5.2
Voltage selection	Configurable from the cloud (one hardware platform, many valves)	DS §5.5.2, §4.10
Source of solenoid power	PoE in the field; USB-C for test/commissioning (refines DS §5.5.2)	FL 7.0/8.0; DS §5.5.2
Connector	Male 3-pin mini-XLR (valve presents female mini-XLR)	FL 1.0; WG
Protection	Current limiting, flyback/free-wheeling, predictable behaviour under open-load and short-circuit	DS §5.5.2
Buffering	Local conditioning so activation does not disturb the supply	DS §5.5.2

The configured solenoid voltage (`SOL_VDC`) is stored on the device and selectable from a provided option list (DS §4.8, §4.10). The PoE power budget for the valve load is the key open item (§7; FL 7.0).

3.9 GPIO & expansion

- **2 × GPIO (DS §5.5.1)**: reserved for future use — IR activation, device lockout, piezo/pushbutton activation, or simulating a push-button for an external controller (Galvin PB2). Where **PWM** is available it is preferred, to allow flow-meter or pressure-sensor inputs.
- **RJ11 expansion port (DS §5.5.4)**: headroom for future development, including moving from a 1:1 to a many-TMV-to-one-device topology. The intended bus to **remote I/O cards** is **BACnet or Modbus** — Modbus preferred where the link should be hidden from third-party access, BACnet where it should be open (FL 5.0). Two architecture questions remain open (§7): whether remote I/O cards carry their own power supply (PoE may not cover them), and how many remote cards attach to one master device.

4. Firmware Specification

The Design Scope draws an explicit line between **device logic** and **cloud logic** in its flow and service-button diagrams (DS §5.1.5–§5.1.8). This section specifies the device side only. Cloud-owned computation — temperature classification (TEMP_CLASS), maintenance-date arithmetic, stagnation counting across groups, alert/email dispatch — is named here purely as the interface the firmware feeds (DS §4.9–§4.13, §5.3–§5.4).

4.1 Firmware role

The transmitter is a **telemetry and execution node** (DS §6.1.2). Safety-critical thresholds (scald limit, stagnation, alert rules) live in the cloud; the firmware measures, computes per-event summaries, applies cloud-pushed configuration, and executes commanded actions. It does not make independent safety decisions.

4.2 Event measurement & packet computation

A packet is emitted to the cloud on each qualifying flow event or service-button press; the cloud extracts its fields into the asset table (DS §4.8). Temperature values are **computed on the device**, with sampling/averaging during flow chosen for accuracy (DS §4.8). Per event the firmware computes (DS §4.6, §4.8):

Field	Definition	Source
MIN_TEMP / MAX_TEMP / AVG_TEMP	Min / max / average temperature, evaluated from 20 s into the flow event	DS §4.6, §4.8
COMP_TEMP	Average of readings at 5 s after the stabilisation period — the compliance reading	DS §4.6, §4.8
PEAK_TIME	Time (s) within the event at which peak temperature occurred	DS §4.8
FLOW_DUR	Total flow-event duration (s)	DS §4.8
SCALD_YN / SCALD_TIME	Scald flag and accumulated seconds above the scald limit (§4.4)	DS §4.8
FS_ERROR	Flow switch open > 2 h (§4.3)	DS §4.8
EVENT_TIME	Event timestamp	DS §4.8

4.3 Flow-event state machine

Gating rules the firmware enforces locally (DS §5.1.1, §3.3, Appendix flow scenarios):

- **Stabilisation (STABLE, default 20 s):** readings in the first STABLE seconds are ignored before data recording. If an event is shorter than STABLE, only flow-duration is sent (DS §5.1.1, §4.11).
- **Flow reject (FLOW_REJECT, default 5 s):** if event duration < FLOW_REJECT, no packet is sent (DS §5.1.1, §4.11).
- **Flow-switch error:** a flow event longer than 2 h raises FS_ERROR (DS §5.1.1, §4.8).
- **Compliance window:** flow longer than 25 s (STABLE + 5 s) yields a captured COMP_TEMP for cloud classification (DS §5.1.5).

STABLE and FLOW_REJECT are cloud-pushed group parameters the device must mirror and update when they change (DS §4.8, §4.11).

4.4 On-device safety & event logic

- **Scald detection (DS §5.1.2):** when measured temperature exceeds TEMP_SCALD, the firmware starts a timer; if it stays above the limit for longer than SCALD_TIME it sets the scald flag in the packet and records the scald duration. The threshold and duration are cloud-defined (DS §4.12); the device only measures and flags.
- **Hot-flush measurement (DS §5.1.4):** the firmware supplies the temperature and duration from which a hot-flush (e.g. > 60 °C for 10 min, or > 70 °C for 5 min) is recognised; the flush-date ledger itself is cloud logic.
- **Service button (DS §5.1.7, §4.6):** a **short (1 s)** press logs general/annual maintenance — Status LED flashes once; a **long (3 s)** press logs 5-year service — Status LED flashes three times. The press event and timestamp are sent; the maintenance *dates* are set in the cloud. The Design Scope flags the 1 s / 3 s scheme as error-prone and asks for a more robust, reversible scheme (DS §5.1.7) — carried to §7.

4.5 Commissioning & calibration

Commissioning is performed on-site over BLE with the CliniMix Setup app, fully offline-capable, storing values on the transmitter until the cloud is reachable (DS §5.2, §7-app). The firmware:

- Exposes the device GUID via the QR linkage for asset binding (DS §6.1.3).
- Stores the **Commissioning Offset (OFFSET)** produced by the calibration wizard: the plumber runs the outlet ≥ 1 min, reads a calibrated reference thermometer, and the app computes the offset that aligns the digital probe to the physical reading (DS §3.2, §7-app, §4.9). **OFFSET** is applied to subsequent readings on-device.
- Accepts the **Operational Performance Setpoint (TEMP_SP)** and **IP configuration** captured during setup (DS §4.9, §4.10).

4.6 Timekeeping

The device maintains an RTC but treats **cloud time as authoritative**; all timestamps are synchronised to cloud date/time, with a configurable time zone (DS §4.1). The RTC preserves ordering and event timing through network or power interruptions.

4.7 Networking & connectivity

- **Addressing:** DHCP by default, static IP configurable via the app (DS §4.7).
- **Architecture:** device direct to cloud over Ethernet; functions without on-site servers or controllers (DS §4.7).
- **App link:** BLE for commissioning; optional Wi-Fi (DS §4.4).

4.8 Offline buffering

When the cloud is unreachable, all event packets and service-button records are **retained locally** and uploaded when connectivity returns. On reaching the storage limit the firmware applies a **first-in, first-out (FIFO)** policy, discarding the oldest record to admit the newest (DS §5.2). Local buffer depth is an open item pending the packet-memory budget (DS §4.1 "Packet Memory: TBD"; §7). The feedback loop also raises **optional MCU SD-card storage** to retain longer sensor-history data on-device (FL 9.0, pending dev); if adopted, it provides storage for both the offline buffer and extended history.

4.9 Security & OTA

- **Encrypted device-to-cloud pathways** within a direct device-to-cloud architecture that removes intermediary servers and reduces attack surface (DS §4.7, §6).
- **Secure over-the-air (OTA) firmware updates**, so fixes and improvements deploy without physical access (DS §4.7, §6).
- **Immutable identity:** the firmware binds permanently to the manufacturer GUID surfaced by the QR code, preserving history across name/location changes and guarding against asset substitution (DS §6.1.3).
- **Constrained local configuration:** because safety parameters are cloud-managed, the device inherently limits unauthorised local change (DS §6.1.2).

4.10 Solenoid control firmware (optional)

Active only when **Flow** or **Both** mode is selected at commissioning (DS §5.1.8). The firmware supports (DS §5.1.8):

- **Automatic activation:** periodic flush (no activation for a set duration), scheduled purge (e.g. 19:00 daily), and stagnation flush (when the group stagnation counter reaches threshold).
- **Input lockout:** physical inputs (piezo/sensor) can be remotely locked from the portal; locked inputs are ignored until cleared.
- **Runtime configuration:** per-activation open duration (**SOL_DUR**) configurable from the Setup Tool or portal (DS §4.9).
- **Usage quotas / anti-vandal:** a maximum number of activations per minute / hour / day; once reached the input is disabled until the window resets.
- **Activation sources:** cloud push, automated logic, or the Setup Tool, even with no physical trigger fitted. Each activation records **SOL_EXEC_TIME**, **SOL_DUR** and **SOL_TRIGG** (DS §4.6, §4.9).

5. Compliance & Standards

Area	Requirement	Source
Emissions	EN IEC 61000-6-3:2021	DS §4.5
Product safety	AS/NZS 62368.1	DS §4.5

Area	Requirement	Source
EMC (radiated/conducted)	AS/NZS CISPR 32	DS §4.5
Regulatory mark	RCM (Australia)	DS §4.3
Radio (BLE / optional Wi-Fi)	AS/NZS 4268	platform target
Additional targets (platform)	CE, FCC, EMC per CISPR 32 / IEC 61000-4-x	platform target
TMV maintenance intervals referenced by logic	AS 4032.3 (annual MAINT_1 = 365 d; 5-yearly MAINT_5 = 1825 d)	DS §4.9, §5.1.6

Exact test plans are to be confirmed with the supplier (DS §4.5). The radio and broader CE/FCC/EMC marks reflect the Nube iO ACX platform certification set.

6. Key Reference Components

Function	Reference part / platform	Notes	Source
Platform	Nube iO ACX (drawing ACX-001 Rev V01)	8 × I/O, PoE/12 VDC, RTC, opt. RS-485/LoRa, IPx5	PO
Application MCU + BLE/ Wi-Fi	Application microcontroller	BLE for Setup app; optional Wi-Fi	DS §4.4; PO
Network + power	100 Mb Ethernet PHY + IEEE 802.3 PoE PD	PoE class per budget incl. valve (§7)	DS §4.1; PO; FL 7.0
Temperature probe	EXSENSE TS103F25C3950FA-ML1300A	10 kΩ NTC, B3950, SS304 NPT1/8, 1300 mm; 2-pin IP65 plug	TS; FL 2.0
Solenoid valve (driven load)	Wallgate WVPB400-2	Inline brass DC solenoid; female mini-XLR lead	WG; FL 1.0
Solenoid power	PoE (field) / USB-C (test)	6–24 V DC configurable	FL 7.0/8.0; DS §5.5.2
Timekeeping	On-board RTC	Cloud-disciplined	DS §4.1

7. Open Items & Engineering Recommendations

Items the Design Scope records as *TBD* or that the live **Hardware Development Feedback Loop (FL)** is tracking, with Nube iO's position and the current status. Resolving these is the purpose of this design cycle; once Galvin's inputs (FL items marked *Pending Client*) are received, Nube iO expects to complete layout and schematic within ~2 weeks (FL 6.0).

#	Open item	Nube iO position / recommendation	Status
1	PoE power budget — must power the valve(s) (DS §4.1; FL 7.0/8.0) — High	Complete the power budget (MCU + PHY + valve drive); confirm a single PoE class covers device + valve actuation, with USB-C carrying enough for commissioning checks. Sizing-critical.	Pending Client
2	Valve connector (DS §4.2; FL 1.0)	Valve presents a female mini-XLR → use a male 3-pin mini-XLR on the device; document pinout/polarity for the 6–24 V range.	Pending Client
3	Temperature-probe connector (DS §4.2; FL 2.0)	Adopt a 2-pin IP65 waterproof plug with strain relief, mating the EXSENSE 1300 mm UL4478 lead (TS §1).	Pending Dev
4	RJ45 connection / waterproofing (DS §4.2; FL 3.0)	Choose among external RJ45 port, IP67 connector, sealed cable, or internal-connect-then-close per the final box design.	Pending Client
5	Product box dimensions & internal layout (DS §4.3; FL 4.0) — High	Need Galvin's detailed cabinet dims and component layout to finalise connector placement and the ACX casing (reconcile DS 60×120×40/IPx4 vs ACX-001 150×40×35/IPx5).	Pending Client
6	Expansion / remote I/O cards (DS §5.5.4; FL 5.0)	Bus is BACnet or Modbus (Modbus to hide from third parties). Confirm whether remote cards self-power and how many attach per master.	Pending Client
7	MCU SD-card / history storage (DS §4.1, §5.2; FL 9.0)	Optional SD card to hold extended sensor history plus the FIFO offline buffer; size for ≥ 30 days of events per asset on non-volatile storage.	Pending Dev
8	Service-button robustness (DS §5.1.7)	Add confirmation feedback (LED + cloud echo) and make service entries reversible by authorised users only; consider an app-confirmed action over bare 1 s / 3 s hold-times.	Nube iO rec.

#	Open item	Nube iO position / recommendation	Status
9	Measurement accuracy budget (DS §5.5.5 vs TS §3)	Lock bias/ADC reference and sense-current so electronics stay < 1 %; state one end-to-end figure covering probe + electronics + self-heating.	Nube iO rec.
10	Enclosure colour / markings & data retention (DS §4.3, §6)	Confirm Galvin Blue PMS-2935 + mark set; device-side retention bounded by the FIFO/SD buffer, long-term retention is a cloud responsibility.	Pending Client

Appendix A — Requirements Traceability

Maps Galvin Design Scope clauses to the responsible domain. **HW / FW** = Nube iO scope (this document); **Cloud** = Galvin/Rubix, interface only.

Design Scope clause	Item	Owner	This doc
§4.1	Power, RTC, environmental, IP	HW	§3.1, §3.3, §4.6
§4.2	Connectors	HW	§3.4
§4.3	Enclosure, LEDs, button, labelling	HW	§3.5
§4.4	Communication (LAN / BLE / Wi-Fi)	HW/FW	§3.2, §4.7
§4.5	Compliance testing	HW	§5
§4.6, §4.8	Data acquisition & packet fields	FW	§4.2
§4.9–§4.13	Cloud parameters / thresholds	Cloud	interface — §4.1
§5.1.1	Flow gating (STABLE, FLOW_REJECT, 2 h)	FW	§4.3
§5.1.2	Scald detection	FW	§4.4
§5.1.3	Flow / low-flow detection	FW	§3.7, §4.3
§5.1.4	Hot-flush measurement	FW (Cloud ledger)	§4.4
§5.1.5	Over-temp / COMP_TEMP capture	FW (Cloud classifies)	§4.3
§5.1.6	Maintenance interval arithmetic	Cloud	interface — §4.4
§5.1.7	Service button	FW	§4.4
§5.1.8	Solenoid & input control	FW	§4.10
§5.2	Offline FIFO buffering	FW	§4.8
§5.3–§5.4	Alerts / reports / email	Cloud	interface — §4 intro
§5.5.1	GPIO / PWM	HW	§3.9
§5.5.2	Solenoid output driver	HW/FW	§3.8, §4.10
§5.5.3	USB-C input / power path	HW	§3.3
§5.5.4	Expansion port (RJ11)	HW	§3.9
§5.5.5	Flow switch & thermistor	HW	§3.6, §3.7
§6	Security, OTA, data sovereignty	FW (Cloud hosting)	§4.9

Appendix B — Thermistor R–T Reference (operating band)

Extract from the EXSENSE datasheet R-T table (TS §6), nominal resistance, covering the device's 0–65 °C operating ambit (DS §4.1) and highlighting the 40–60 °C mixed-water band.

Temp (°C)	R nominal (kΩ)	Temp (°C)	R nominal (kΩ)
0	31.898	40	5.338
5	25.074	45	4.373
10	19.739	50	3.591
15	15.665	55	2.977
20	12.480	60	2.477
25	10.000	65	2.078

Temp (°C)	R nominal (kΩ)	Temp (°C)	R nominal (kΩ)
30	8.075	—	—
35	6.538	—	—

Appendix C — Definitions (device-relevant subset)

TMV Thermostatic Mixing Valve · **PoE** Power over Ethernet · **USB-PD** USB Power Delivery · **RTC** Real-Time Clock · **NTC** Negative Temperature Coefficient · **GPIO** General-Purpose Input/Output · **PWM** Pulse-Width Modulation · **OTA** Over-the-Air · **DHCP** Dynamic Host Configuration Protocol · **FIFO** First-In, First-Out · **GUID** Globally Unique Identifier · **BLE** Bluetooth Low Energy · **LAN** Local Area Network · **RCM** Regulatory Compliance Mark (Australia). (Source: DS §8.)